

ORIGINAL CONTRIBUTION

Cerebral Edema in Patients With Large Ischemic Core After Thrombectomy: A Secondary Analysis of SELECT2 Randomized Trial

Felix Ng¹, PhD; Vignan Yogendrakumar², MD; Hannah Johns³, PhD; Leonid Churilov⁴, PhD; Ameer Hassan⁵, DO; Michael Abraham⁶, MD; Santiago Ortega-Gutierrez⁷, MD; M. Shazam Hussain, MD; Michael Chen⁸, MD; Clark W. Sitton⁹, MD; Scott E. Kasner¹⁰, MD; Gagan Sharma, MCA; Prodipta Guha¹¹, BSc; Deep Pujara¹², MBBS; Vitor Mendes Pereira¹³, MD; Maarten G. Lansberg¹⁴, MD; Lawrence Wechsler¹⁵, MD; Thanh N. Nguyen¹⁶, MD; Johanna T. Fifi¹⁷, MD; Michael D. Hill¹⁸, MD; Marc Ribo¹⁹, MD; Mark W. Parsons²⁰, PhD; Stephen M. Davis²¹, MD; James Grotta²², MD; Gregory W. Albers²³, MD; Bruce C.V. Campbell²⁴, PhD*; Amrou Sarraj²⁵, MD*; for the SELECT2 Investigators

BACKGROUND: Cerebral edema is a life-threatening complication of ischemic stroke that disproportionately affects patients with large established infarction. We assessed whether endovascular thrombectomy (EVT) reduces or exacerbates edema development, and the association between edema with short-term and long-term outcomes in this high-risk population.

METHODS: In this prespecified secondary analysis of the SELECT2 randomized clinical trial (Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke), which tested the efficacy and safety of EVT versus medical management in adult patients with acute anterior circulation large vessel occlusion presenting with large ischemic core (defined as Alberta Stroke Program Early Computed Tomography Score of 3–5 or core volume ≥ 50 mL on computed tomography perfusion or diffusion magnetic resonance imaging), we assessed maximum midline shift (MLS) within 7 days of randomization between treatment groups using a probabilistic index model.

RESULTS: After exclusion of 10 patients who underwent hemicraniectomy before follow-up imaging, 342 patients were analyzed. The median MLS on follow-up magnetic resonance imaging or computed tomography was 6.39 mm (interquartile range, 0–12.0) in the EVT and 4.18 mm (interquartile range, 0–9.66) in medical management patients ($P=0.021$). EVT was independently associated with greater MLS (adjusted odds ratio, 1.63 [95% CI, 1.25–2.12]; $P=0.0027$) after adjusting for age and core volume. There was no interaction between EVT and core volume at presentation on the association with MLS ($P>0.79$). MLS was associated with the development of early neurological worsening (adjusted odds ratio, 1.15 [95% CI, 1.07–1.23]; $P<0.001$), and a lower likelihood of long-term functional improvement assessed on modified Rankin Scale score at 90 days (adjusted odds ratio, 0.96 [95% CI, 0.93–0.98]; $P=0.0029$). On mediation analysis, cerebral edema reduced the effect of EVT on functional outcome by 10.6%.

CONCLUSIONS: Despite overall clinical benefit, EVT in patients presenting with a large ischemic core was associated with increased cerebral edema, which, in turn, was associated with early neurological worsening and worse functional outcome at 90 days. Early recognition and individualized treatment to prevent secondary injury from cerebral edema in this population are warranted.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT03876457.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: brain edema ■ ischemic stroke ■ perfusion ■ thrombectomy ■ tomography

Correspondence to: Amrou Sarraj, MD, Department of Neurology, University Hospitals Cleveland Medical Center, Case Western Reserve University, 11100 Euclid Ave, Hanna House No. 504, Cleveland, OH 44106. Email amrou.sarraj@uhhospitals.org

*B.C.V. Campbell and A. Sarraj contributed equally.

This manuscript was sent to Jean-Claude Baron, Guest Editor, for review by expert referees, editorial decision, and final disposition.

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/STROKEAHA.125.054015>.

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Nonstandard Abbreviations and Acronyms

aGenORs	adjusted generalized odds ratios
ANGEL-ASPECT	Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core
CHARM	Intravenous Glibenclamide for Cerebral Edema After Large Hemispheric Stroke
CT	computed tomography
CTP	computed tomography perfusion
ENW	early neurological worsening
EVT	Endovascular thrombectomy
MARVEL	Methylprednisolone as Adjunctive Therapy for Acute Large Vessel Occlusion: A Randomized Double-Blind, Placebo-Controlled Trial in Patients Undergoing Endovascular Therapy
MLS	midline shift
MM	medical management
MR CLEAN	Multicenter Randomized Clinical Trial of Endovascular Treatment for Acute Ischemic Stroke in the Netherlands
NIHSS	National Institutes of Health Stroke Scale
PH	parenchymal hematoma
SELECT2	Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke

Six randomized clinical trials testing the efficacy and safety of endovascular thrombectomy (EVT) in patients presenting with large ischemic stroke have broadly demonstrated superior clinical outcome observed up to 12 months.^{1–7} Despite the demonstrable long-term benefit, early neurological worsening (ENW) within 24 hours posttreatment was more frequent in patients with EVT in the SELECT2 trial (Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke). ENW occurred without a corresponding increase in symptomatic intracranial hemorrhage or procedural complications, and was associated with worse functional outcome at 90 days.⁵ Better understanding and potential treatment of acute clinical deterioration shortly after EVT may augment the clinical benefit of reperfusion.

Cerebral edema, a devastating complication of large infarction with high morbidity and mortality, represents a potential explanation for reperfusion-related ENW in patients presenting with a large ischemic core.^{8,9} Recent

analyses have suggested a complex relationship between reperfusion and cerebral edema that is dependent on the physiological state of ischemic tissue at the time of treatment, whereby reperfusion reduces infarct growth in penumbra tissue but increases mass effect in nonsalvageable tissue.^{10,11} However, these studies primarily included patients with small to moderate core volume, and therefore were underpowered to assess the proposed bimodal effect in patients with larger core volumes. Studying the effects of reperfusion treatment in patients presenting with large core could provide further insight into this hypothesis. To this end, patients with a considerable volume of nonsalvageable tissue are the cohort most susceptible to the effects of cerebral edema and the cohort that would likely benefit most from adjunctive antiedema treatment. To date, the only evidence-based definitive treatment for cerebral edema in stroke is early decompressive craniectomy to reduce the risk of death and severe disability.¹²

We aimed to assess the association between EVT and subsequent cerebral edema development, and the association between edema with short-term and long-term outcomes in the SELECT2 international randomized clinical trial that enrolled patients presenting with a large ischemic core.



METHODS

Study Population

The SELECT2 trial was an international, multicenter, randomized clinical trial with blinded outcome assessment across 31 centers in North America, Europe, Australia, and New Zealand. Adults aged 18 years to 85 years old presenting within 24 hours of last known well with an ischemic stroke due to an occlusion of the internal carotid artery or M1 segment of the middle cerebral artery and a large ischemic core, defined as Alberta Stroke Program Early Computed Tomography (CT) Score of 3 to 5, or core volume ≥ 50 mL on CT perfusion (CTP; tissue with relative cerebral blood flow $<30\%$) or diffusion magnetic resonance imaging (tissue with apparent diffusion coefficient $<620 \times 10^{-6}$ mm²/s) were enrolled and randomized using a Web-based centralized system with covariate adaptive randomization to receive EVT with best medical management (MM) or best MM alone. All patients received the best MM in accordance with local national guidelines. Decisions to perform decompressive hemicraniectomy were made in accordance with local practices and by individual patients' representatives and their clinical teams.¹³ The detailed trial protocol, primary results, and statistical analysis have been published.^{5,14} All participating centers obtained approval from local institutional review boards or equivalent ethics committees. All patients or their surrogate decision maker provided written informed consent. This study followed the Consolidated Standards of Reporting Trials reporting guideline.

Imaging Evaluation

All patients underwent an imaging protocol before enrollment that consisted of a noncontrast CT, CT/magnetic resonance

angiography, and CTP/magnetic resonance diffusion-perfusion imaging. To best estimate the extent of ischemic injury, baseline ischemic core volume in this analysis was assessed as the larger of the CT hypodense lesion (segmented by manual planimetry by maximal visual extent) and CTP lesion (region with relative cerebral blood flow <30%) or the magnetic resonance diffusion-perfusion volume (tissue with apparent diffusion coefficient <620×10⁻⁶ mm²/s). As an alternative marker of nonsalvageable tissue at baseline, the volume of severe hypodensity on the noncontrast CT at presentation with a Hounsfield Unit ≤26 representing substantial blood-brain barrier disruption was derived as per previously published methods.¹⁵ Prespecified follow-up imaging was performed within 24 hours to 7 days of randomization using magnetic resonance imaging or CT. Midline shift (MLS) was determined as the maximum deviation of the septum pellucidum or equivalent midline structures. Symptomatic intracranial hemorrhage was defined as parenchymal hematoma (PH) type II with a deterioration in National Institutes of Health Stroke Scale (NIHSS) score ≥4 compared with the baseline NIHSS score before randomization. Imaging analyses were performed blinded to clinical data and randomization allocation.

Outcomes

The primary analysis assessed maximum MLS on all available imaging within the prespecified follow-up period (up to 7 days of randomization) as the outcome, in accordance with previous published methodologies in EVT trials.¹¹ Patients who underwent hemicraniectomy before imaging could be acquired during the prespecified follow-up period were excluded from the primary analysis. Secondary clinical outcomes were ordinal analysis of the modified Rankin Scale score at 90-day follow-up, and ENW was defined as a ≥4-point increase on the NIHSS score within 24 hours attributed to the stroke rather than another cause, such as hemorrhage, medication effects, or metabolic derangements.

Statistical Analysis

A probabilistic index model using an as-treated approach was used to evaluate treatment effects for the primary outcome adjusted for age and ischemic core volume at baseline. The nonparametric treatment effect estimates were reported using adjusted generalized odds ratios (aGenORs) and 95% CIs, as MLS, analyzed as a continuous variable, was not normally distributed. Interaction between EVT and ischemic core volume on MLS (primary outcome) was tested to determine whether EVT treatment effect on MLS was modified by ischemic core volume. Sensitivity analyses were performed for the following: (1) alternative definitions of ischemic core volume using non-contrast CT hypodense lesion volume segmented by manual planimetry, automated CTP lesion volume or volume of severe CT hypodensity as defined by ≤26 Hounsfield Unit; (2) inclusion of patients with early (<24 hours) hemicraniectomy by imputing a MLS value greater than the highest recorded MLS; (3) inclusion of additional variables in serum blood glucose at randomization and time from randomization to follow-up imaging as continuous variables, and presence of PH on follow-up as a dichotomized variable; and (4) alternative time epochs for follow-up scan where only imaging up to 96 or 48 hours post-randomization was analyzed.

Probabilistic index model in the secondary analysis was performed with ties split equally between groups for ordinal outcomes for the modified Rankin Scale score and adjusted odds ratio with 95% CIs for ENW as a dichotomous variable. To test whether the occurrence of ENW may be better explained by accounting for cerebral edema, we compared the Bayesian Information Criterion of the multivariable models for ENW (as the dichotomous outcome) with and without MLS as a variable.¹⁶ A lower Bayesian Information Criterion by 2 to 6 units is regarded as positive, 6 to 10 as strong, and >10 as very strong evidence of superior model fit.

To investigate and quantify whether the effect of EVT on cerebral edema may reduce the effect of EVT on functional outcome, we conducted a causal mediation analysis using cerebral edema as a mediating variable between EVT and functional outcome, adjusting for age and core volume.¹⁷ Causal mediation analysis decomposes the effect of an intervention into a direct effect (effect of intervention on outcome) and average causal mediated effect (effect of intervention on a mediator, which then has an effect on outcome). For this analysis, we defined the mediator as the presence of cerebral edema defined as MLS of at least 1 mm on axial scan on follow-up CT/magnetic resonance imaging or decompressive hemicraniectomy,¹⁸ and the outcome as modified Rankin Scale score 0 to 3 at 90 days.

Analyses were performed using Stata version 17 (StataCorp) and R version 4.2.2 (The R Foundation) using 2-sided statistical tests, where a *P* < 0.05 was considered significant. Missing data were not imputed apart from the sensitivity analysis that included patients with early hemicraniectomy.

RESULTS

After exclusion of 10 patients with hemicraniectomy (EVT *n*=6; MM *n*=4), 342 out of 352 enrolled patients were analyzed (Table 1; Table S1). The median time to follow-up scan for analysis of MLS was 33.2 hours (20.4–60.5) postrandomization for the EVT group and 26.6 (18.7–45.7) for the MM group. Hemorrhagic transformation types 1 and 2, defined as petechial hemorrhage without mass effect, were more common in the EVT group. The rate of PH was low in both groups (EVT *n*=5, MM *n*=2).

The median ischemic core volume using the larger of the CT hypodensity region or the CTP cerebral blood flow–based lesion was 103 mL (interquartile range [IQR], 69.3–140) in the EVT group and 99.4 mL (74.4–136) in the MM group. The maximum MLS in patients with EVT was 6.39 mm (IQR, 0–12.0) and 4.18 mm (IQR, 0–9.66) in MM patients (*P*=0.021). The timing of the imaging when maximal MLS was recorded did not differ significantly between the treatment arms (EVT 33.2 hours from randomization versus MM 27.5 hours; *P*=0.104). A scatterplot of MLS demonstrating greater MLS in the EVT group across the spectrum of core volume is shown in the Figure. The proportion of patients with radiological evidence of malignant MCA infarction, defined as ≥5 mm MLS^{19,20} was significantly higher in patients with EVT (EVT 56.4 versus 43.6%; *P*=0.021).

Table 1. Patient Characteristics and Outcome of Patients Analyzed

	EVT (n=172)	MM (n=170)	P value
Baseline characteristics			
Age, y	66.5 [59.0–75.0]	67.0 [58.0–75.0]	0.691
NIHSS score	18.5 [15.0–23.0]	19.0 [15.0–22.0]	0.956
Time from last known well to randomization, h	9.17 [5.25–15.4]	9.43 [5.78–15.0]	0.688
Core volume, mL			
CBF	70.0 [39.0–110]	77.0 [50.0–104]	0.451
NCCT	82.6 [46.5–114]	86.2 [49.4–112]	0.811
Largest of CBF and NCCT	100 [68.7–138]	99.9 [73.7–136]	0.865
Volume of severe hypodensity on NCCT with ≤ 26 Hounsfield Units	12.8 [4.70–26.3]	14.4 [5.22–30.8]	0.196
ASEPCTS on baseline CT	4.00 [3.00–5.00]	4.00 [4.00–5.00]	0.685
Serum glucose level, mg/dL	126 [111–158]	129 [112–159]	0.839
Patients with IV thrombolysis, n (%)	34 (19.8%)	28 (16.5%)	0.528
Patients with angiographic reperfusion (eTICI 2b-3), n (%)	136 (79.1%)	...	...
Outcomes			
Maximum MLS across 7 d, mm	6.39 [0–12.0]	4.18 [0–9.66]	0.021
Time of imaging with maximum MLS recorded, h	33.2 (20.4–62.8)	27.5 (19.3–49.8)	0.104
Maximum MLS across 96 h, mm	4.80 [0–8.72]	0 [0–6.21]	0.003
Maximum MLS across 48 h, mm	3.95 [0–7.90]	0 [0–5.46]	0.003
Early neurological worsening, n (%)	42 (24.4%)	26 (15.3%)	0.058
Parenchymal hematoma, n (%)	5 (2.9%)	3 (1.8%)	0.724

Patient characteristics and outcomes of patients included in the primary analysis. Patients who underwent hemicraniectomy before imaging could be acquired during the prespecified follow-up period were excluded from the primary analysis. ASEPCTS indicates Alberta Stroke Program Early Computed Tomography Score; CBF, cerebral blood flow; CT, computed tomography; EVT, endovascular thrombectomy; IV, intravenous; MLS, midline shift; MM, medical management; NCCT, noncontrast computed tomography; and NIHSS, National Institutes of Health Stroke Scale.

Thrombectomy Treatment Effect on MLS

EVT was independently associated with greater MLS (aGenOR, 1.63 [95% CI, 1.25–2.12]; $P=0.0027$) after adjusting for age and core volume at baseline. There was no interaction between EVT and core volume on the association with MLS ($P=0.79$). Sensitivity analyses testing (1) alternative definitions of ischemic core volume, (2) inclusion of hemicraniectomy patients, (3) different time epochs of follow-up scan, and (4) inclusion of additional adjusting covariates (time of follow-up scan analyzed, presence of PH, and baseline glucose level) yielded concordant results (Table 2).

Conversely, after adjusting for EVT, core volume (aGenOR, 1.02 [95% CI, 1.01–1.02]; $P<0.001$) and glucose at presentation (aGenOR, 1.00 [95% CI, 1.00–1.01]; $P=0.01$) were associated with MLS (Table 3). Baseline perfusion mismatch ratio at presentation was not associated with MLS when analyzed instead of core volume (aGenOR, 0.80 [95% CI, 0.54–1.20]; $P=0.28$; Table S2). Among EVT patients, angiographic reperfusion was not associated with MLS (aGenOR, 0.92 [95% CI, 0.55–1.54]; $P=0.75$).

Association of MLS With Clinical Outcome

Patients with ENW within 24 hours had greater MLS (5.05 mm [IQR, 2.06–10.20] versus 0.00 mm [IQR,

0.003–3.94]; $P<0.001$). MLS was associated with the development of ENW (adjusted odds ratio, 1.15 [95% CI 1.07–1.23]; $P<0.001$) when adjusting for core volume, EVT, age, and baseline NIHSS score. The Bayesian Information Criterion was 15.6 units lower when MLS was included, indicating very strong evidence of improved model fit (Table 4).

At 90 days, MLS was associated with worse functional outcome (aGenOR, 0.96 [95% CI 0.93–0.98]; $P=0.0024$) when adjusting for core volume, EVT, age, and baseline NIHSS score. EVT was associated with better functional outcome (aGenOR adjusted odds ratio, 1.93 [95% CI, 1.45–2.58]; $P<0.001$) when adjusting for MLS, age, and baseline NIHSS score.

Mediation Analysis

The average direct effect of EVT on achieving a functional outcome of modified Rankin Scale score of 0 to 3 (risk difference, 0.217 [95% CI, 0.133–0.298]; $P<0.001$) was reduced by 10.1% as a result of increased chance of MLS on functional outcome (average causal mediated effect, -0.022 [95% CI, -0.048 to -0.0003]; $P=0.015$) for a total effect risk difference of 0.195 ([95% CI, 0.111–0.278]; $P<0.001$).

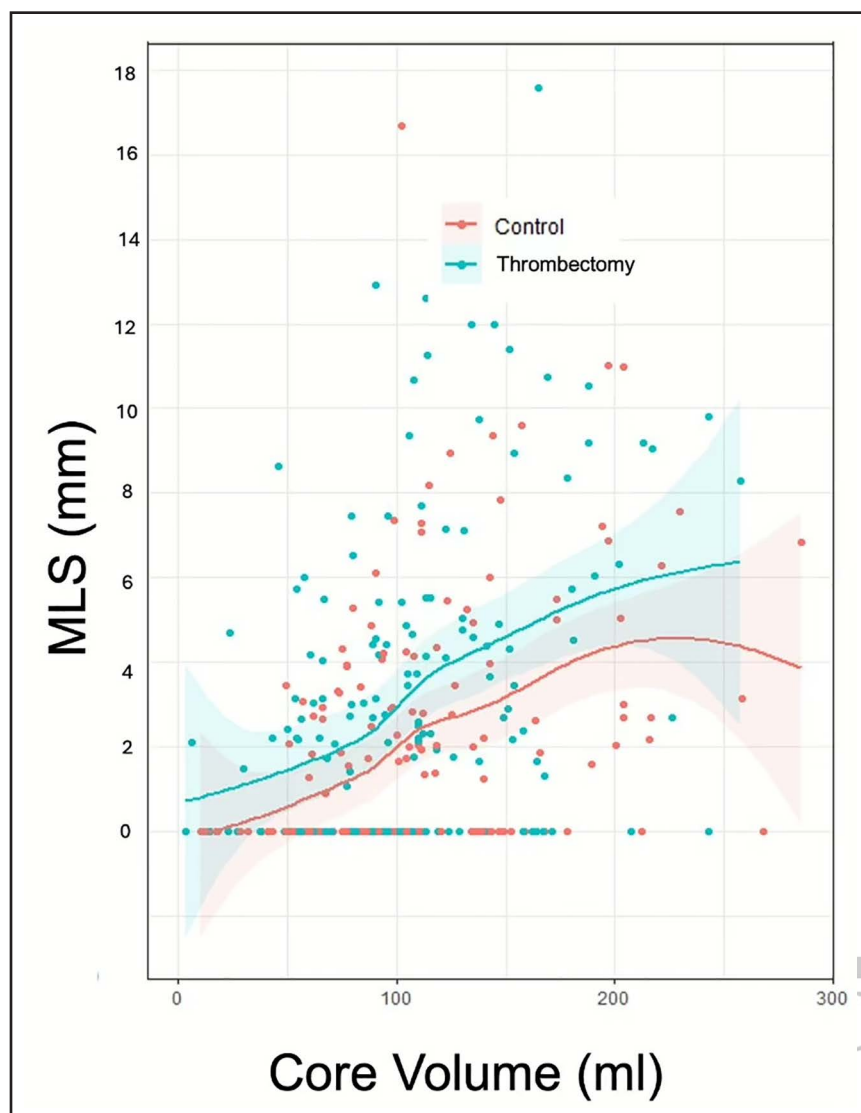


Figure . Scatterplot of maximum midline shift (MLS) across baseline ischemic core volume, along with locally weighted scatterplot smoothing lines and corresponding 95% CI.

Patients who received thrombectomy had greater cerebral edema as assessed by MLS across the full range of ischemic core volume.



DISCUSSION

In a secondary analysis of reperfusion treatment effect on edema development in an international randomized clinical trial of patients presenting with large ischemic core, EVT was consistently associated with more edema as characterized by MLS on follow-up across the range of ischemic core volume. Cerebral edema was associated with worse neurological outcome shortly after EVT and provides an explanation for early clinical deterioration in patients receiving EVT. Overall, cerebral edema modestly reduced the clinical benefit of EVT.

Earlier studies have appeared to show conflicting results on the effect of reperfusion on cerebral edema development. Preclinical studies demonstrated reperfusion to exacerbate edema, whereas human data showed reperfusion to reduce edema instead.^{21–27} By extrapolating pooled randomized controlled trial data primarily of patients with small to moderate core volume, more recent clinical analyses proposed a unifying explanation

by demonstrating a variable tissue response to reperfusion that is dependent on the balance between penumbra salvage with infarct growth reduction versus reperfusion-induced swelling of infarcted tissue.^{10,11,18} The current analysis provides important novel insights and strong support for this hypothesis by demonstrating an increase in edema after reperfusion treatment in a prospective cohort of patients enrolled for the presence of a large ischemic core. This increase in edema was independent of age, core volume, follow-up scan time, baseline glucose level and PH. Specifically, we adjusted for the presence of PH to account for potential influences in MLS measurements related to increased PH in the EVT group. A higher rate of petechial hemorrhage was observed in the EVT group. As petechial hemorrhage, by definition of the Heidelberg bleeding classification, does not exert substantial mass effect, it is felt unlikely to confound MLS measurements, and an association between petechial hemorrhage and edema more likely results from the common pathophysiology

Table 2. Association of Thrombectomy With Cerebral Edema

Multivariable model with EVT as independent variable	Outcome	aGenOR (95% CI)	P value
Covariates: age, core volume	Maximum MLS over 7 d	1.63 (1.25–2.12)	0.0027
Sensitivity analysis—alternative core definition			
Covariates: age, core volume (NCCT)	Maximum MLS over 7 d	1.56 (1.20–2.03)	0.0082
Covariates: age, core volume (CBF)	Maximum MLS over 7 d	1.52 (1.17–1.97)	0.0020
Covariates: age, core volume (volume of severe hypodensity on NCCT with ≤ 26 HU)	Maximum MLS over 7 d	1.48 (1.14–1.92)	0.0032
Sensitivity analysis—including patients with hemicraniectomy			
Covariates: age, core volume	Maximum MLS over 7 d	1.48 (1.15–1.90)	0.0025
Sensitivity analysis—including additional variables			
Covariates: age, core volume, randomization to follow-up scan time, baseline glucose level, presence of PH on follow-up	Maximum MLS over 7 d	1.60 (1.22–2.10)	<0.0001
Sensitivity analysis—alternative time epoch			
Covariates: age, core volume	Maximum MLS over 48 h	1.61 (1.22–2.12)	0.0067
Covariates: age, core volume	Maximum MLS over 96 h	1.68 (1.29–2.20)	0.0013

In the primary analysis, when comparing 2 patients with similar age and core volume, the odds of developing a greater MLS are increased by 59% if the patient had a thrombectomy. Sensitivity analyses were performed for (1) alternative definitions of ischemic core volume in noncontrast CT hypodense lesion volume, or CT perfusion lesion volume; (2) inclusion of patients with early (<24 h) hemicraniectomy by imputing a MLS value greater than the highest recorded MLS; (3) inclusion of additional variables in serum blood glucose at randomization and time from randomization to follow-up imaging as continuous variables, and presence of parenchymal hematoma on follow-up as a dichotomized variable; (4) alternative time epochs for follow-up scan where only imaging up to 96 or 48 h postrandomization. aGenOR indicates adjusted generalized odds ratio; CBF, cerebral blood flow; CT, computed tomography; EVT, endovascular thrombectomy; HU, Hounsfield Unit; MLS, midline shift; NCCT, noncontrast CT; and PH, parenchymal hematoma.



of blood-brain barrier injury.^{16,28} As larger infarcts have more blood-brain barrier disruption, codevelopment of edema and petechial hemorrhage after reperfusion is well described.^{29,30} In a retrospective study of 840 patients with infarct involving $\geq 1/2$ of the MCA territory on postintervention imaging, more patients with petechial hemorrhage had ≥ 5 mm MLS or underwent decompressive craniectomy than patients without hemorrhagic transformation (46.9% versus 17.0%).³⁰ Similarly, a secondary analysis on edema progression of the ANGEL-ASPECT (Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core) randomized trial observed more ICH of any type in the EVT group (49.6% versus 16.8%).³¹

A key question relating to this hypothesized bimodal reperfusion-edema response is whether there is a core volume or a threshold of the nonsalvageable tissue at which the relationship inflects. We observed a consistent

detrimental treatment effect on edema development regardless of core volume in the SELECT2 cohort. This was observed across the alternative definitions of core volume, which included assessment by the extent of hypodensity on noncontrast CT as a marker of blood-brain barrier disruption at a tissue level and by the degree of critical hypoperfusion on CTP.^{15,32,33} However, an earlier exploratory analysis of the 7 randomized controlled trials comparing thrombectomy versus medical therapy in patients pooled in the HERMES collaboration that primarily enrolled based on the presence of salvageable tissue with limited core volume suggested 130 mL as the baseline core volume above which edema was exacerbated.¹¹ Future analyses comprising a balanced pool of patients representing the full range of core volumes encountered in clinical practice will better characterize when the detrimental effect of reperfusion-induced edema supercedes the protective effect of penumbra

Table 3. Baseline Variables Associated With Cerebral Edema

Variables	aGenOR	95% CI	P value
Age (for every 1-y increment)	1.01	0.99–1.02	0.35
Core volume (for every 1-mL increment)	1.02	1.01–1.02	<0.001
Time to randomization from onset (for every 1-h increment)	1.02	1.00–1.05	0.07
Glucose at presentation (for every 1 mg/dL increment)	1.00	1.00–1.01	0.02
Thrombectomy (compared with medical care)	1.60	1.22–2.10	0.001
Parenchymal hematoma	1.88	0.62–5.70	0.26

PIM was used to test the effects of baseline variables on cerebral edema as characterized by midline shift, adjusting for thrombectomy and parenchymal hematoma on follow-up in an exploratory multivariable model. The effect estimates were expressed as aGenORs. Core volume and glucose at presentation were associated with worse cerebral edema. aGenOR indicates adjusted Generalized odds ratio; and PIM, probabilistic index model.

Table 4. Association of Cerebral Edema With ENW

Multivariable model	Outcome	BIC	ΔBIC
Variables: MLS, age, core volume, EVT, baseline NIHSS score	ENW	297.78	15.60
Variables: age, core volume, EVT, baseline NIHSS score	ENW	313.39	

The BIC of the multivariable models for ENW (as the dichotomous outcome) with MLS was 297.78 and 313.39 without MLS as a variable. A decrease in BIC by >10 indicates very strong evidence of superior model fit attributable to the inclusion of the variable in the multivariable model. BIC indicates Bayesian Information Criterion; ENW, early neurological worsening; EVT, endovascular thrombectomy; MLS, midline shift; and NIHSS, National Institutes of Health Stroke Scale.

salvage and reduced infarct growth. Moreover, recent studies have shown that the traditional dichotomy of infarcted versus noninfarcted brain is an oversimplification, as the severity of injury within the conventionally defined infarct lesion may vary.^{32,34} Incorporating further assessment of the extent of ischemic injury at baseline within the core lesion by imaging surrogates, such as more directly quantifying the degree of Blood-brain barrier disruption, will provide additional understanding into the impact of reperfusion on cerebral edema.^{16,35}

In the SELECT2 trial, patients receiving EVT had better outcomes at 90 days but experienced more early deterioration that was not accounted for by peri-procedural complications or PH. To date, the cause of ENW remains unclear.⁸ In this analysis, we demonstrated that cerebral edema was associated with ENW and was a determinant of ENW in the multivariable model tested. Collectively, our findings suggest that the overall benefit of EVT may be partially compromised by EVT-related cerebral edema in patients presenting with large infarction, and therefore, reducing cerebral edema may further augment the long-term benefit of EVT in this population. This is supported by a secondary analysis of the ANGEL-ASPECT trial, which found an early increase in MLS following EVT at 24 hours partially mediated poor functional outcome after EVT.³¹ In this context, despite overall neutral trial results, the phase 3 CHARM randomized controlled trial (Intravenous Glibenclamide for Cerebral Edema After Large Hemispheric Stroke) showed potential benefit of intravenous glibenclamide in the subgroup of patients with moderately large ischemic core (<120 mL) who underwent EVT.³⁶ A secondary analysis of the MARVEL trial (Methylprednisolone as Adjunctive Therapy for Acute Large Vessel Occlusion: A Randomized Double-Blind, Placebo-Controlled Trial in Patients Undergoing Endovascular Therapy) showed that in patients with intracranial ICA occlusion, use of methylprednisolone was associated with better 90-day independent ambulation and a lower rate of hemicraniectomy use.³⁷ Further prospective clinical studies will be required to confirm these findings. The immediate implication of our findings for clinical practice is to consider the additional risk of reperfusion-related edema early when treating patients presenting with a large ischemic core with regard to the potential need for surgical decompression or osmotic therapies, especially among patients with risk factors of cerebral edema, such as late presenters with hyperglycemia. Baseline

imaging biomarkers such as collateral status, hypoperfusion intensity ratio, and net water uptake may further risk-stratify patients susceptible to reperfusion-related edema and warrant future studies.

Limitations

Our study has several limitations. The SELECT2 trial was not designed to assess the treatment effect on cerebral edema. Our findings should therefore be regarded as exploratory. Second, as the trial protocol allowed flexibility for follow-up imaging to be performed over 7 days, there was heterogeneity in the timing of follow-up scans between patients. Accordingly, we analyzed the scan with the highest MLS if multiple scans were acquired to capture incidences of clinical deterioration where patients may have required an additional scan due to evolving cerebral edema. A previous analysis from the MR CLEAN trial (Multi-center Randomized Clinical Trial of Endovascular Treatment for Acute Ischemic Stroke in the Netherlands) found concordant results when assessing the relationship of EVT relationship between early (24 hours) and late (5–7 days) follow-up imaging.²⁴ We have further performed sensitivity analyses for different follow-up scan time epochs and found similar results. Third, although MLS is the widely accepted reference standard for quantifying cerebral edema, it is predominantly a marker of mass effect rather than a direct measure of fluid influx. Although alternative measures of cerebral edema, such as net water uptake, are able to infer pathological tissue water uptake, this imaging biomarker is known to be compromised by hemorrhagic transformation and occult residual contrast staining of the infarct postthrombectomy and cannot be generalized to magnetic resonance imaging, which limits its application to the current data set.^{38,39} Fourth, treatment of cerebral edema was according to local standards; hence, differences in treatment practices may have affected our results.

Conclusions

In this prespecified analysis of the SELECT 2 prospective randomized clinical trial that enrolled patients presenting with a large ischemic core, reperfusion treatment with EVT was associated with increased cerebral

edema development, in contrast to prior studies based on patients with large penumbra and small ischemic core. This study confirms a complex bimodal relationship between reperfusion treatment and edema. Accordingly, cerebral edema contributed to the occurrence of ENW and worse outcome at 90 days among patients treated with EVT. Early treatment and proactive approaches targeting cerebral edema in this population may further improve the benefit of EVT.

ARTICLE INFORMATION

Received October 3, 2025; final revision received February 28, 2026; accepted March 12, 2026.

Affiliations

Department of Neurology, Melbourne Brain Centre at the Royal Melbourne Hospital, Parkville, Australia (F.N., V.Y., G.S., P.G., S.M.D., B.C.V.C.). Department of Medicine, University of Melbourne, Parkville, Australia (H.J., L.C.). University of Texas Rio Grande Valley–Valley Baptist Medical Center, Harlingen (A.H.). Department of Neurology and Radiology, University of Kansas Medical Center (M.A.). Department of Neurology, University of Iowa (S.O.G.). Department of Neurology, Cleveland Clinic, OH (M.S.H.). Department of Neurosurgery, Rush University Medical Center, Chicago, IL (M.C.). Department of Diagnostic and Interventional Radiology, UTHealth McGovern Medical School, Houston, TX (C.W.S.). Division of Vascular Neurology, University of Pennsylvania, Philadelphia (S.E.K.). Department of Neurology, Case Western Reserve University, Cleveland, OH (D.P., A.S.). Division of Neurology, St. Michael's Hospital, Toronto, Ontario, Canada (V.M.P.). Department of Neurology, Stanford University, CA (M.G.L., G.W.A.). Department of Neurology, Hospital of the University of Pennsylvania, Philadelphia (L.W.). Department of Neurology, Boston Medical Center, MA (T.N.N.). Department of Neurology, Icahn School of Medicine at Mount Sinai, New York, NY (J.T.F.). Department of Clinical Neurosciences, University of Calgary, Alberta, Canada (M.D.H.). Department of Neurology, Hospital Vall d'Hebrón, Barcelona, Spain (M.R.). Department of Neurology, University of Newcastle, New South Wales, Australia (M.W.P.). Department of Neurology, Memorial Hermann Hospital Texas Medical Center, Houston (J.G.).

Sources of Funding

SELECT2 (Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke) was supported by an investigator-initiated grant from Stryker Neurovascular to University Hospitals Cleveland Medical Center and the University of Texas McGovern Medical School. Dr Ng is supported by National Health and Medical Research Council of Australia (2022/GNT2018497).

Disclosures

Dr Ng reports grants from National Health and Medical Research Council and grants from Heart Foundation. A. Hassan reports compensation from Medtronic for consultant services; an ownership stake in Quantanosis.ai; compensation from viz.ai for consultant services; employment by Valley Baptist Medical Center; compensation from Penumbra Inc for consultant services; compensation from GE Healthcare for consultant services; compensation from Cerenovus for consultant services; compensation from MicroVention Inc for consultant services; and compensation from Stryker Corporation for consultant services. Dr Abraham reports compensation from Imperative Care Inc for consultant services and compensation from Rapid Medical Ltd for consultant services. Dr Ortega-Gutierrez reports grants from National Institutes of Health; grants from Siemens; compensation from Medtronic for consultant services; stock options in brainflow; grants from PCORi; compensation from MicroVention Inc for consultant services; stock options in Active Motif; grants from methinks; stock holdings in Eureka Therapeutics; grants from National Institutes of Health; grants from Stryker; compensation from Stryker for consultant services. Dr Hussain reports compensation from Stryker for data and safety monitoring services. Dr Chen reports compensation from Route 92 Medical Inc for consultant services; compensation from Medtronic for consultant services; compensation from Vesalio for consultant services; compensation from MicroVention Inc for consultant services; compensation from Q'apel for consultant services; compensation from Penumbra Inc for consultant services; compensation from RapidPulse for consultant services; compensation from Stryker Corporation for consultant services; and compensation from Johnson & Johnson Health Care Systems Inc. for consultant services. Dr Sitton reports compensation

from Case Western Reserve University for consultant services. Dr Kasner reports employment by Bayer. Dr Mendes Pereira reports compensation from Stryker for consultant services and compensation from Medtronic Inc for consultant services. Dr Wechsler reports compensation from SELECT LATE for data and safety monitoring services; compensation from Takeda Pharmaceuticals for data and safety monitoring services; compensation from Athersys for consultant services; compensation from Annals of Neurology for other services; compensation from Astellas Pharma for consultant services; compensation from Silver Creek Pharmaceuticals for consultant services; compensation from Aruna Bio for consultant services; compensation from Brainsgate for consultant services; compensation from Novartis for consultant services; and stock options in Forest Devices. Dr Nguyen reports compensation from the American Stroke Association for other services; compensation from Route 92 Medical Inc for consultant services; compensation from Kaneka for other services; compensation from Genentech for other services; and compensation from Bayer for consultant services. Dr Fifi reports employment by Mount Sinai Health System; compensation from Medtronic for consultant services; stock holdings in Precision Recovery; compensation from MIVI for data and safety monitoring services; stock holdings in Radical; stock holdings in Q'apel; compensation from MicroVention Inc for consultant services; compensation from Cerenovus for consultant services; compensation from Stryker Corporation for consultant services; stock holdings in Imperative Care Inc; and compensation from Siemens Healthcare for consultant services. Dr Hill reports grants from NoNO Inc; grants from Boehringer Ingelheim; grants from Canadian Institutes of Health Research; grants from Medtronic; compensation from Diamedica Inc for consultant services; and stock options in Basking Bioscience LLC. Dr Ribo reports compensation from Stryker Corporation for consultant services; stock holdings in Nora; compensation from AptaTargets for consultant services; compensation from Vesalio for consultant services; stock holdings in Methinks; compensation from Philips for consultant services; stock holdings in Anaconda Biomed; compensation from Rapid Pulse for consultant services; compensation from Medtronic MiniMed Inc for consultant services; compensation from Cerenovus for consultant services; and compensation from Senseo for data and safety monitoring services. Dr Davis reports compensation from Amgen for other services. Dr Albers reports compensation from iSchemaView for consultant services and stock holdings in iSchemaView. Dr Sarraj reports grants from Stryker; and grants from Patient-Centered Outcomes Research Institute. The other authors report no conflicts.

Supplemental Material

Tables S1–S2
CONSORT Checklist

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